Why a single binary operator deserves attention

A speculative companion to the EML formalization

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Abstract

This is a deliberately speculative companion to the technical formalization paper (`notes/extended_paper/eml_for`
The formalization establishes that one binary operator, $\mathrm{eml}(x, y) := \exp(x) - \log(y)$, paired with the constant 1, generates the 36-element catalogue of elementary functions used in typical scientific calculators [1]. The purpose of *this* document is to argue that the existence of such a continuous Sheffer operator is not a curiosity of the elementary-function algebra alone: similar "one binary plus one terminal" generative reductions arise across several distinct areas of physics — some explicitly, some implicitly — and the EML observation can be used to clarify what structural fact is being relied on in each case.

We make no claim of formal isomorphism; the analogies offered here are structural prompts, not theorems. The reader is invited to take them as a programme of *questions* for the relevant sub-communities, rather than as established connections.

Contents

1 Scope and disclaimer

This note is exploratory. The technical paper [2] establishes a precise mathematical fact (a Lean 4 + Mathlib formalization of Odrzywołek's completeness theorem); the present document is concerned with whether that fact points to a broader pattern. We separate the two so that the technical paper can stand on its formal-methods merits without being diluted by speculative analogies.

The four sections that follow each pose the question *"in domain X, is there a structural reduction analogous to the EML Sheffer operator?"* and offer a sketch of what an affirmative answer would look like. Sections 2 (Hamiltonian normal forms), 3 (Wess–Zumino covering identities), 4 (conformal-bootstrap reductions), and 5 (single-instruction analog hardware) are each a few pages of motivated speculation. Section 6 collects the open problems that fall out of the analogies.

2 Hamiltonian normal forms and the Sheffer reduction

[TO BE WRITTEN — develops the analogy between (a) the EML reduction of the 36-primitive elementary toolbox to a single binary operator and (b) the reduction of a generic Hamiltonian system to its Birkhoff–Gustavson normal form, where a complicated dynamical system collapses to a finite expression in a small canonical generator set. The point is: in both cases the existence of a small generative basis is non-trivial and has computational value.

Open question: does the Birkhoff–Gustavson normal form admit a "Sheffer reduction" — a single canonical-transform kernel from which all integrable normal forms can be recovered, paired with a finite seed? If yes, what is the kernel?]

3 Wess–Zumino-style covering identities

[TO BE WRITTEN — discusses the 4d Wess–Zumino superalgebra's generator reduction (the entire algebra is built from a single odd spinor charge Q_α and its conjugate, plus the bracket; analogue to "one binary plus one terminal"). Comments on whether Sheffer-style reductions exist for the bosonic generators of the super-Poincaré algebra.

Open question: is the supercharge's generative role essential, or is it a notational convenience? In the EML case we can prove (via the formalization) that no smaller generator set works for the full 36-primitive catalogue; what is the analogous statement in the supercharge setting?]

4 Gauge-fixing and conformal-bootstrap reductions

[TO BE WRITTEN — Belavin–Polyakov–Zamolodchikov's conformal bootstrap reduces the infinite-dimensional space of CFT correlators to a small set of structure constants and conformal blocks; the analogue is a Sheffer-style reduction in the algebra of correlators.

Open question: in the bootstrap algebra, is the OPE the "single binary operator" analogue, with the identity operator playing the role of the constant 1? If so, what is the analogue of the 36-primitive catalogue — the list of correlators that the bootstrap is required to reproduce?]

5 Single-instruction analog computing

[TO BE WRITTEN — discusses FPGAs, photonic integrated circuits, and the practical implication of the EML Sheffer: a "single-instruction math co-processor" that evaluates only $\mathrm{eml}(a, b)$ in hardware. The formal proof of completeness implies that such a co-processor is universal for the 36-primitive class, modulo the boundary points; this is a substantially stronger statement than the empirical verification provides.

Open question: what is the silicon-area / latency / energy tradeoff of a single-EML-instruction analog co-processor, compared with the standard floating-point unit that implements all 36 primitives separately? At what circuit size does the Sheffer reduction become economically interesting?]

6 Open questions

The four analogies above each pose a concrete open question (highlighted at the end of the corresponding section). Collected here for ease of reference:

1. **Birkhoff–Gustavson kernel.** Does the Birkhoff–Gustavson normal form admit a single canonical-transform kernel from which all integrable normal forms can be recovered, paired with a finite seed?

2. **Supercharge minimality.** In the 4d Wess–Zumino algebra, can the role of the spinor supercharge as "the single generator" be made into a formal Sheffer-style minimality theorem?

3. **OPE as Sheffer operator.** In the conformal bootstrap, can the operator product expansion be cast as a Sheffer operator on the algebra of correlators, with the identity operator as constant?

4. **Single-EML hardware.** What is the practical viability of a single-instruction EML analog co-processor for elementary function evaluation?

7 What the formalization tells us

[TO BE WRITTEN — closing observation. The formalization makes two contributions to the broader programme: (a) it establishes that *the EML Sheffer reduction is a theorem, not a conjecture*, so analogues in physics can be discussed without the loophole "but is it really minimal?"; (b) the witness-family form (one term per region of the input domain) is precisely the form that a single-instruction hardware unit would naturally implement, since hardware does case analysis on input range as a matter of course.

The technical paper restricts itself to the formalization. This note exists to argue that the formalization's value extends beyond formal methods: a structural fact about elementary functions is also a structural prompt for several adjacent fields, and the prompt is now mathematically settled.]

References

[1] A. Odrzywołek, *All elementary functions from a single binary operator*, arXiv:2603.21852, 2026. `https://arxiv.org/abs/2603.21852`

[2] B. Naskrecki, *Formalizing the EML completeness theorem in Lean 4: an extended account of the construction and of the multi-agent workflow that produced it*, companion paper `notes/extended_paper/eml_formalization_paper.pdf` in the eml-formalization repository, 2026.